

2D Arrays

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Notes

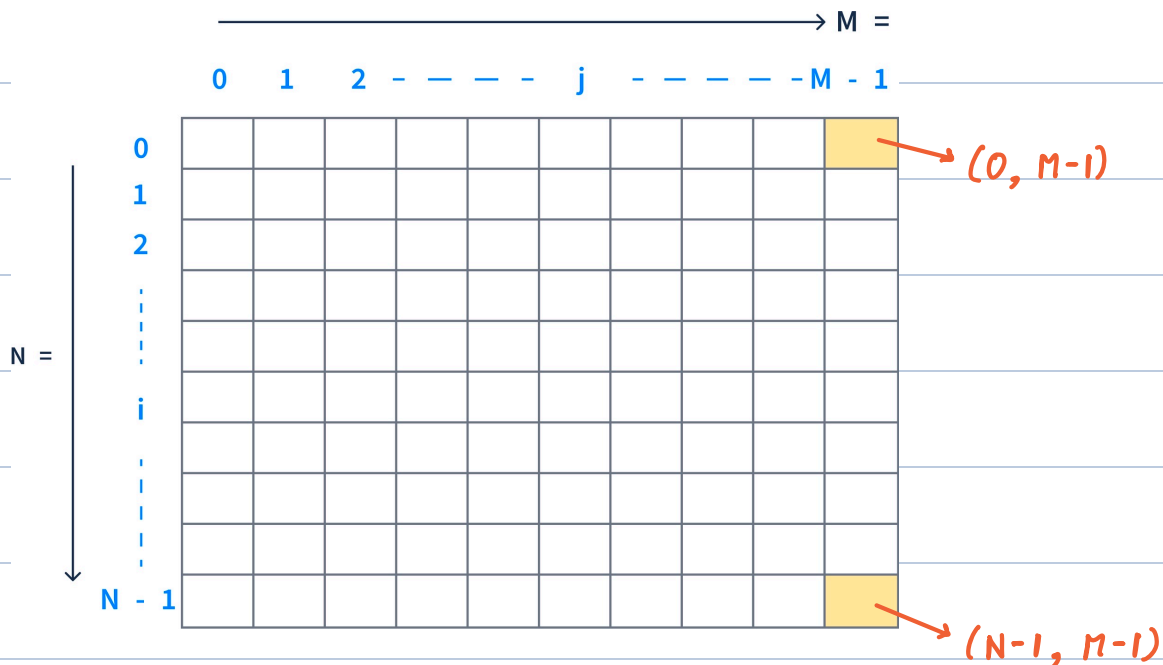
	0	1	2	
0	(0,0)	(0,1)	(0,2)	Column Index
1	(1,0)	(1,1)	(1,2)	
2	(2,0)	(2,1)	(2,2)	
				Row Index

0 ————— (N-1)





- `int arr[ N ][ M ]` , total number of elements $\rightarrow$



How is it actually stored?

$A[i] \rightarrow \text{Base address} + (i * \text{datatype size})$

$A[i][j] \rightarrow \text{Base address} + (i * M + j) * \text{datatype size}$



< Question > : Given $\text{arr}[N][M]$, print row-wise sum.

• $\text{arr}[3][4] \rightarrow$

	0	1	2	3	<u>o/p</u>
0	10	2	7	3	22
1	9	5	-1	8	21
2	3	11	15	20	49

```
for i → 0 to (N-1) {  
    sum = 0  
    for j → 0 to (M-1) {  
        sum += A[i][j]  
    }  
    print (sum)  
}
```

$TC = \underline{O(N * M)}$ $SC = \underline{O(1)}$



< **Question** > : Given $\text{arr}[N][M]$, print column-wise sum.

• $\text{arr}[3][4] \rightarrow$

	0	1	2	3
0	10	2	7	3
1	9	5	-1	8
2	3	11	15	20

o/p $\rightarrow$ 22 18 21 31

```
for i  $\rightarrow$  0 to (M-1) {  
    sum = 0  
    for j  $\rightarrow$  0 to (N-1) {  
        sum += A[j][i]  
    }  
    print(sum)  
}
```

$TC = O(N * M)$

$SC = O(1)$



< Question > : Given $\text{arr}[N][N]$ $N * N \rightarrow$ Square matrix

Principal Diagonal

	0	1	2	3
0	1	5	8	7
1	2	11	3	9
2	15	20	-3	18
3	30	40	50	60

4*4

i	j
0	0
1	1
2	2
3	3

```
for i → 0 to (N-1) {
    | print (A[i][i])
}
```

TC = $O(N)$

SC = $O(1)$

Anti Diagonal

	0	1	2	3
0	1	5	8	7
1	2	11	3	9
2	15	20	-3	18
3	30	40	50	60

4*4

i	j
0	3
1	2
2	1
3	0

$i + j = \text{constant}$

$i + j = N - 1$

$$j = N - 1 - i$$

```
for i → 0 to (N-1) {
    | j = N - 1 - i
    | print (A[i][j])
}
```

TC = $O(N)$

SC = $O(1)$



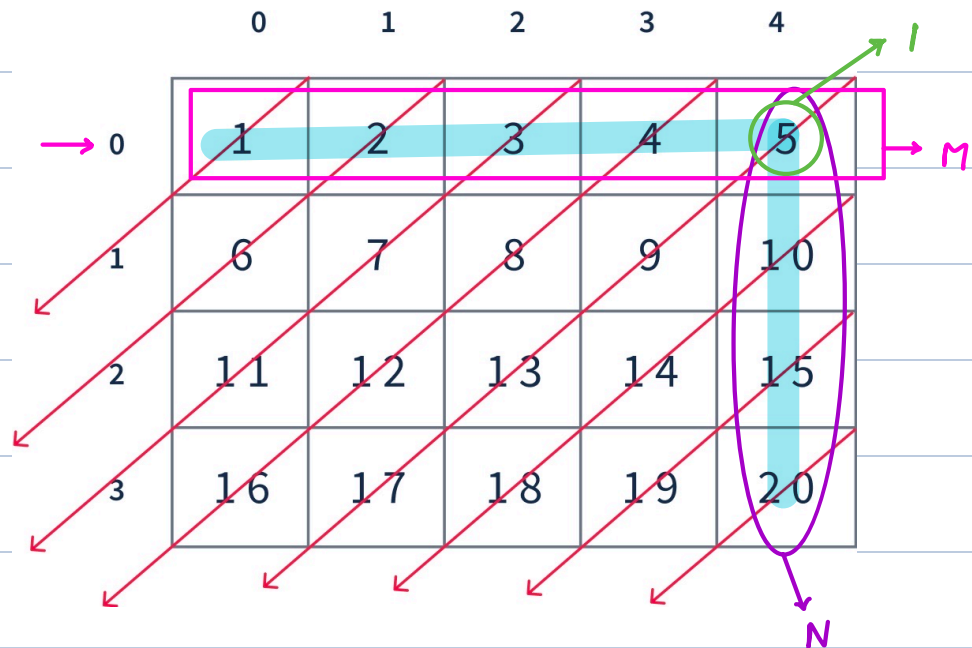
< Question > : Given $\text{arr}[N][M]$. Print all the elements diagonally from right to left.

• $\text{arr}[4][5] \rightarrow$

o/p $\rightarrow$

```

1
2 6
3 7 11
4 8 12 16
5 9 13 17
10 14 18
15 19
20
  
```



diagonals = $N + M - 1$

$r++$
 $c--$

```

for i  $\rightarrow$  0 to (M-1) {
    r = 0    c = i
    while (r < N && c >= 0) {
        print (A[r][c])
        r++    c--
    }
    print ("\n")
}
  
```



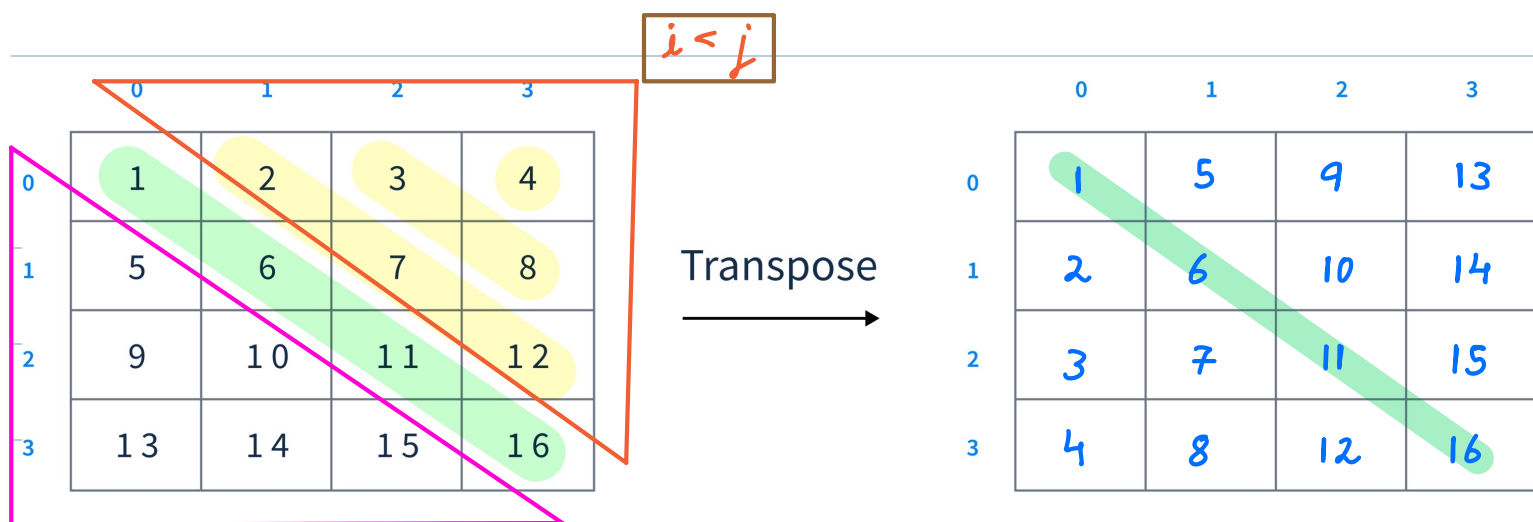
```
for i → 1 to (N-1) {  
    r = i    c = M-1  
    while (r < N && c >= 0) {  
        print (A[r][c])  
        r++    c--  
    }  
    print ("\\n")  
}
```

$$TC = \underline{O(N * M)}$$

$$SC = \underline{O(1)}$$



Transpose of a Square Matrix



$i > j$

row $\longleftrightarrow$ col

$i, j \longleftrightarrow j, i$

for $i \rightarrow 0$ to $(N-1)$ {

for $j \rightarrow (i+1)$ to $(N-1)$ {

$t = A[i][j]$

$A[i][j] = A[j][i]$

$A[j][i] = t$

} swap

}

}

TC = $O(N^2)$

SC = $O(1)$



Rotate a mat[N][N]

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12
3	13	14	15	16

Rotate by
90° Clockwise

	0	1	2	3
0	13	9	5	1
1	14	10	6	2
2	15	11	7	3
3	16	12	8	4

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

90°

13	9	5	1
14	10	6	2
15	11	7	3
16	12	8	4

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12
3	13	14	15	16

Transpose

	0	1	2	3
0	1	5	9	13
1	2	6	10	14
2	3	7	11	15
3	4	8	12	16

Transpose + Reverse every row → Rotated A[3][3]

```

for i → 0 to (N-1) {
    for j → (i+1) to (N-1) {
        t = A[i][j]
        A[i][j] = A[j][i]
        A[j][i] = t
    }
}

```

} swap

```

for i → 0 to (N-1) {
    l = 0    r = M-1
    while (l < r) {
        t = A[i][l]
        A[i][l] = A[i][r]
        A[i][r] = t
        l++    r--
    }
}

```

$$TC = O(N^2 + N^2) = \underline{O(N^2)}$$

$$SC = \underline{O(1)}$$

